UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 002 · Nernst equation and its derivation · compiled 17 October 2026

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1 Short notes

1 · Free energy and electrode potential

The maximum work obtainable equals the decrease in free energy, $-\Delta G = W_{max}$. For an electrochemical system, W_{max} = total charge (nF, $F = 96500$ C mol^{-1}) × energy per unit charge (E), so $W_{max} = nFE$, $\Delta G = -nFE$ and $\Delta G^\circ = -nFE^\circ$. A positive emf means negative ΔG: a spontaneous reaction.

2 · Derivation of the Nernst equation for a single electrode

For $\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}$: $\Delta G = \Delta G^\circ + RT\ln Q$ with $Q = [\mathrm{M}]/[\mathrm{M^{n+}}]$. Substituting gives $-nFE = -nFE^\circ + RT\ln\frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$; dividing by $-nF$ and putting $[\mathrm{M}] = 1$ gives $E = E^\circ + \frac{RT}{nF}\ln[\mathrm{M^{n+}}]$. With $\ln = 2.303\log$ at 298 K: $E = E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$.

3 · Using the single-electrode equation: concentration and temperature

E rises by $0.0591/n$ V per tenfold rise in $[\mathrm{M^{n+}}]$ and equals E° at 1 M. Zn in 0.01 M $\mathrm{Zn^{2+}}$: −0.8191 V; Cu in 0.1 M $\mathrm{Cu^{2+}}$: +0.3105 V. The factor $2.303RT/F$ is proportional to T: 0.0591 V at 298 K, 0.0611 V at 308 K.

4 · Nernst equation for a cell

For $a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$: $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$, n from the balanced reaction, solids omitted. ESA Jan 2025: 2Al + 3$\mathrm{Cu^{2+}}$(0.6 M) → 2$\mathrm{Al^{3+}}$(0.35 M) + 3Cu, $n = 6$, $Q = 0.5671$, $E_{cell} = 2.0024$ V.

2 Keyword index

17 terms; the number in brackets is the topic.

0.0591 V (3). $2.303RT/F$ at 298 K.

2.303 (2). $\ln 10$: converts $\ln$ to $\log$.

Balanced reaction (4). Needed for n and the powers in Q.

Driving force (1). E_{cell} drives the cell reaction.

Energy per unit charge (1). The electrode potential E (J C^{-1} = V).

Equilibrium constant (4). $\log K = nE^{\circ}_{cell}/0.0591$ (added for clarity).

Faraday constant (1). 96500 C per mole of electrons.

Free energy change (1). $\Delta G = -nFE$; negative for a spontaneous reaction.

Maximum work (1). $W_{max} = -\Delta G = nFE$.

Nernst equation (2). Relation between electrode potential and concentration of the species the electrode is reversible with.

Pure substance (2). Activity 1, so $[\mathrm{M}] = 1$.

Reaction quotient (2). Q = products/reactants with coefficients as powers.

Slope 0.0591/n (3). Change of E per decade of concentration.

Standard electrode potential (2). E° at 1 M, 1 atm, 298 K.

Standard emf (4). $E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{anode}$.

Temperature effect (3). Factor $2.303RT/F$ grows with T (0.0611 V at 308 K).

Total charge available (1). nF.

3 Rapid recall

1	Define the Nernst equation.	Relation between electrode potential and the concentration of the species the electrode is reversible with.
2	Maximum work in terms of free energy?	$W_{max} = -\Delta G$.
3	Maximum electrical work?	nFE (charge nF × energy per unit charge E).
4	ΔG and E?	$\Delta G = -nFE$; $\Delta G^\circ = -nFE^\circ$.
5	Thermodynamic equation used?	$\Delta G = \Delta G^\circ + RT \ln Q$.
6	Q for $\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}$?	$[\mathrm{M}]/[\mathrm{M^{n+}}]$.
7	Why is $[\mathrm{M}] = 1$?	Pure solid.
8	Divide by what?	$-nF$.
9	Nernst equation, single electrode, 298 K?	$E = E^\circ + \frac{0.0591}{n} \log[\mathrm{M^{n+}}]$.
10	Value of $2.303RT/F$ at 298 K?	0.0591 V (0.05913 with slide constants).
11	At 308 K?	0.0611 V.
12	Nernst equation for a cell?	$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log Q$.
13	Zn in 0.01 M $\mathrm{Zn^{2+}}$ ($E^\circ = -0.76$ V)?	−0.8191 V.
14	ESA Al–Cu cell: n, Q, emf?	6; 0.5671; 2.0024 V.
15	Sign of E for a spontaneous reaction?	Positive ($\Delta G < 0$).
16	Slope of E vs $\log[\mathrm{M^{n+}}]$?	$0.0591/n$.

Mnemonic (study aid). *"Work, Gibbs, Quotient, Divide, One, Log."* The six moves of the derivation: $W_{max} = nFE \to \Delta G = -nFE \to +RT \ln Q \to \div(-nF) \to [\mathrm{M}] = 1 \to$ 2.303 log at 298 K.

4 Last 10 minutes before the exam

Top facts

1. $-\Delta G = W_{max} = nFE$; $\Delta G^\circ = -nFE^\circ$.
2. $\Delta G = \Delta G^\circ + RT\ln Q$ (no nF here).
3. Electrode: $Q = [\mathrm{M}]/[\mathrm{M^{n+}}]$, $[\mathrm{M}] = 1$.
4. $E = E^\circ + \frac{RT}{nF}\ln[\mathrm{M^{n+}}] = E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$ at 298 K.
5. $2.303 \times 8.314 \times 298/96500 = 0.0591$ V; 0.0611 V at 308 K.
6. Cell: $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log\frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$.
7. n from the balanced equation; solids out of Q.
8. ESA Jan 2025 Al–Cu: $n = 6$, $Q = 0.5671$, $E_{cell} = 2.0024$ V.
9. E changes by $0.0591/n$ V per tenfold concentration change.
10. $E > 0 \Leftrightarrow \Delta G < 0 \Leftrightarrow$ spontaneous.

Top formulas

E1.2 · Maximum electrical work

ESA ★★★

$$W_{max} = \underbrace{nF}_{\text{charge}} \times \underbrace{E}_{\text{energy/charge}} = nFE$$

GIVES	Electrical work from n mol of electrons at potential E.
APPLIES	Any electrode or cell.
UNITS	C mol^{-1} × V = J mol^{-1}
NOTE	1 V = 1 J C^{-1}.

S002·3 · T001 00:20:21 · T001 00:21:06

E1.3 · Free energy and potential

ESA ★★★

$$\Delta G = -nFE, \qquad \Delta G^\circ = -nFE^\circ$$

GIVES	Links thermodynamics to potential.
APPLIES	Second form: standard conditions (1 M, 1 atm, 298 K).
UNITS	J mol^{-1}
NOTE	$E > 0 \Rightarrow \Delta G < 0$: spontaneous.

S002·3 · T001 00:16:25 · T001 00:21:24

E2.1 · Free energy and reaction quotient

ESA ★★★

$$\Delta G = \Delta G^\circ + RT \ln Q$$

GIVES How ΔG changes as Q changes.

APPLIES Any reaction; $\ln$ is the natural log.

UNITS J mol^{-1}

NOTE No nF here: $RT \ln Q$, not $\frac{RT}{nF} \ln Q$.

S002·4 · T001 00:21:42

E2.3 · After substituting and dividing by −nF

ESA ★★★

$$-nFE = -nFE^\circ + RT \ln \frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]} \implies E = E^\circ - \frac{RT}{nF} \ln \frac{[\mathrm{M}]}{[\mathrm{M^{n+}}]}$$

GIVES The general Nernst equation, before simplification.

APPLIES Any T.

UNITS V

NOTE Dividing by $-nF$ flips the sign of the RT term.

S002·4 · T001 00:22:32 · T001 00:23:01

E2.5 · Single electrode at 298 K

ESA ★★★

$$E = E^\circ + \frac{0.0591}{n} \log[\mathrm{M^{n+}}]$$

GIVES The form used in numericals.

APPLIES 298 K; reduction potential.

UNITS V

NOTE $0.0591 = 2.303 \times \frac{8.314 \times 298}{96500}$.

S002·5 · T001 00:23:29 · T001 00:23:50

E4.1 · Nernst equation for a cell (298 K)

ESA ★★★

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$$

GIVES Cell emf at non-standard concentrations.

APPLIES $a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$; n from the balanced equation.

UNITS V

NOTE General T: replace 0.0591 by $2.303RT/F$.

S002·5 · T001 00:24:15 · T001 00:25:06